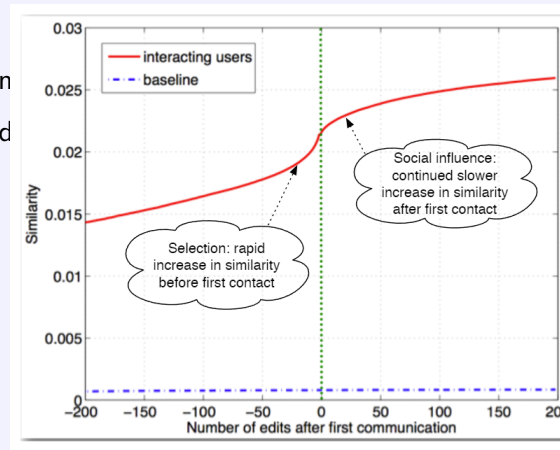


Selection and Social Influence: Editing Wikipedia

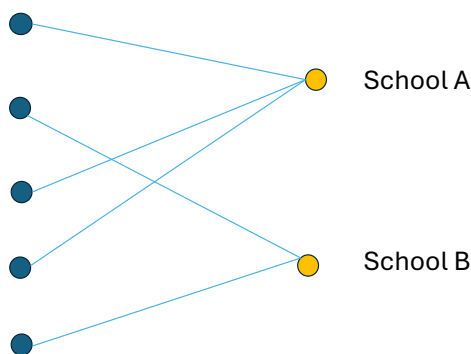
- This graph (Crandall et.al. 2008) plots the average similarity between two editors of Wikipedia before and after their first contact (at 0).
- It shows that similarity goes up both before and after the first contact (selection and influence).



Affiliation Network

- For homophily, we introduced heterogeneity of nodes.
- Another type of information we can add to nodes is their affiliation/membership.
- We can create **affiliation network** by introducing two kinds of nodes: people and groups/institutions, and two kinds of links: people-people links and affiliation links.

Example: schools and student (Bipartite graph)



This graph represents which student attends to which school (so only affiliation links). Such two-sided graph where nodes in one side are only connected to the nodes on the other side is called **bipartite graph**.

Triadic Closure in Affiliation Networks

In affiliation networks, we can consider richer patterns of triadic closures.

There are three patterns.

- If you are a friend of A and B , then A and B tend to be friends (usual one).
- If two persons are affiliated with the same organization, they tend to be friends.
- If you are affiliated with some organization, then your friend tends to be affiliated with the same organization.

Summary

- Homophily: more connections among similar people than the theoretical values.
- There are different mechanisms that cause homophily: (a) selection, (b) selection via covariates, and (c) social influence.
- In reality, both selection and social influence is at work. But it is difficult to determine which is more important.
- Affiliation network is a related type of networks where people may be linked to people and/or organizations.

Suggested reading. EK Ch.4.1-4.4